JAWAHARLAL COLLEGE OF ENGINEERING & TECHNOLOGY
JAWAHAR GARDENS, LAKKIDI MANGALAM, PALAKKAD
(An ISO 9001:2015 Certified Institution)

DEPARTMENT OF COMPUTER SCIENCE (CYBERSECURITY)

BLOCKCHAIN TECHNOLOGY FOR CYBERSECURITY

ABSTRACT

Blockchain technology, originally developed to support cryptocurrencies, has significant potential for improving cybersecurity and protecting digital information. A blockchain is a distributed ledger in which transactions or records are stored across multiple participating systems and protected using cryptographic techniques. This seminar explores how blockchain concepts can be applied to address cybersecurity challenges such as data tampering, identity theft, unauthorized modification, and lack of transparency. Because blockchain records are designed to be difficult to alter without detection, the technology can provide an additional layer of integrity for sensitive digital records. The seminar examines applications such as decentralized identity management, secure data sharing, software integrity verification, access control, and audit logging. It also discusses how smart contracts can automate security-related processes while reducing dependence on centralized authorities. However, blockchain is not a complete solution to cybersecurity problems, and challenges such as scalability, privacy, key management, implementation complexity, and smart-contract vulnerabilities must be considered. The seminar provides an overview of both the security benefits and limitations of blockchain technology and evaluates its potential role in future cybersecurity systems. By combining cryptography, decentralization, and distributed verification, blockchain can offer new approaches to protecting digital assets and strengthening trust in online environments.

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